

Abelacimab: A leap forward in anticoagulation with FXI and FXIa Inhibition.

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Direct Oral Anticoagulants (DOACs) have revolutionized the treatment of thromboembolic disorders, offering targeted, effective, and safer alternatives to traditional anticoagulants like heparins and vitamin K antagonists (VKAs). Despite their benefits, DOACs have drawbacks, including an increased risk of gastrointestinal bleeding and unsuitability for patients with mechanical heart valves. Recent research has highlighted Factor XI (FXI) as a promising anticoagulation target due to its significant role in pathological thrombosis and minor involvement in normal hemostasis. Abelacimab, an antibody that inhibits FXI, has shown potential in transforming anticoagulation therapy by sparing hemostasis. This review provides a comprehensive analysis of abelacimab, examining its clinical pharmacology and its pharmacokinetic and pharmacodynamic properties. It scrutinizes abelacimab's safety profile and key monitoring parameters. The current evidence supporting its use and potential future research strengthening its position in anticoagulant therapy is also discussed. The objective is to enhance understanding and contribute to discussions around developing safer anticoagulants, particularly for patients at risk for thrombosis.

Drug-Drug Interactions of FXI Inhibitors: Clinical Relevance.

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Inhibitors of the factor FXI represent a new class of anticoagulant agents that are facing clinical approval for the treatment of acute coronary syndrome (ACS), venous thromboembolism (VTE), and stroke prevention of atrial fibrillation (AF). These new inhibitors include chemical small molecules (asundexian and milvexian), monoclonal antibodies (abelacimab, osocimab, and xisomab), and antisense oligonucleotides (IONIS-FXIRX and fesomersen), and thus, they have very peculiar and different pharmacokinetic and pharmacodynamic properties. Besides their clinical efficacy and safety, based on their pharmacological heterogeneity, the use of these drugs in patients with comorbidities may undergo drug-drug interactions (DDIs) with other concomitant therapies. Although only little clinical evidence is available, it is possible to predict clinically relevant DDI by taking into consideration their pharmacokinetic properties, such as the CYP450-dependent metabolism, the interaction with drug transporters, and/or the route of elimination. These characteristics may be useful to differentiate their use with the direct oral anticoagulant (DOAC) anti-FXa (rivaroxaban, apixaban, edoxaban) and thrombin (dabigatran), whose pharmacokinetics are strongly dependent from P-gp inhibitors/inducers. In the present review, we summarize the current clinical evidence on DDIs of new anti FXI with CYP450/P-gp inhibitors and inducers and indicate potential differences with DOAC anti FXa.

Factor XI inhibition in cardiovascular disease.

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Cardiovascular disease remains the leading global cause of mortality, requiring effective antithrombotic strategies to prevent thromboembolic events. Currently available therapies are effective but have inherent bleeding risks which may limit or preclude their use, particularly in patients at the highest risk of bleeding. Factor XI (FXI) inhibitors are a promising new class of anticoagulants which may mitigate the risk of bleeding while maintaining efficacy. Further, they have the potential to provide effective

anticoagulation in indications where direct oral anticoagulants (DOACs) are proven less effective than vitamin K antagonists (VKAs) or when DOACs are contraindicated. The development of FXI inhibitors was based on mechanistic considerations suggesting FXI's role in thrombus formation without significantly affecting hemostasis, supported by epidemiological data and animal experiments. FXI inhibitors, including antisense oligonucleotides, monoclonal antibodies, and small-molecule inhibitors, target different stages of FXI production or activation, offering a diversity of therapeutic options with differing onset and offset of action, drug interaction potential, and renal elimination. FXI inhibitors have shown potential benefits in phase II trials, demonstrating similar or reduced bleeding rates to existing agents, including DOACs. The early termination of AZALEA-TIMI 71 (Safety and Tolerability of Abrelximab [MAA868] vs Rivaroxaban in Patients With Atrial Fibrillation) and OCEANIC-AF (A Study to Learn How Well the Study Treatment Asundexian Works and How Safe it is Compared to Apixaban to Prevent Stroke or Systemic Embolism in People With Irregular and Often Rapid Heartbeat [Atrial Fibrillation], and at Risk for Stroke) trials underscores challenges in the selection of appropriate patient populations and anticoagulant class, agent, and dose. Ongoing phase III trials including OCEANIC-STROKE (A Study to Test Asundexian for Preventing a Stroke Caused by a Clot in Participants After an Acute Ischemic Stroke or After a High-risk Transient Ischemic Attack, a So-called Mini Stroke) and LIBREXIA trials aim to further explore the efficacy of FXI inhibitors in stroke, acute coronary syndrome, and atrial fibrillation. In conclusion, FXI inhibitors hold promise as next-generation anticoagulants, potentially addressing limitations of current therapies. Ongoing research is required to establish their place in clinical practice and address unresolved questions.

New targets for antithrombotic medications: seeking to decouple thrombosis from hemostasis.

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Arterial and venous thromboses are the leading causes of morbidity and mortality worldwide. Numerous antithrombotic agents are currently available with antiplatelet, thrombolytic/fibrinolytic, and anticoagulant activity. However, all the currently available antithrombotic agents carry a risk of bleeding that often prevents their use. This unfavorable risk-benefit profile is particularly challenging for patients with cancer-associated venous thromboembolism, patients with atrial fibrillation at a high risk of bleeding, and patients with end-stage renal disease. Patients with ischemic stroke and acute coronary syndromes have not yet found a favorable risk-benefit profile with anticoagulant therapy to help reduce the residual thromboembolic risk that remains after antiplatelet and lipid therapy. Two emerging classes of antithrombotic agents, factor (F)XI or activated factor XI (FXIa) inhibitors and glycoprotein VI inhibitors, have shown promise in their ability to prevent pathologic thrombosis without increasing the risk of hemostatic-related bleeding in phase 2 studies. Among the FXI/FXIa inhibitors of coagulation, a parenterally administered monoclonal antibody (abelximab) and 2 orally administered small molecule inhibitors (asundexian, milvexian) are collectively being studied in patients with atrial fibrillation, cancer-associated venous thromboembolism, acute coronary syndrome, and ischemic stroke. One parenterally administered glycoprotein VI antiplatelet agent (glenzocimab) is currently being studied in patients with ischemic stroke. If shown to be efficacious and safe in ongoing phase 3 studies, both classes of emerging antithrombotic agents have the potential to greatly improve outcomes for patients with challenging thrombotic conditions.

The factor XI/XIa antibody abelacimab combined with enoxaparin inhibits filter clotting in hemodialysis circuits ex vivo.

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Drugs targeting factor XI may offer an alternative to heparin for preventing blood clotting in extracorporeal circulation. We investigated the effects of abelacimab, a novel monoclonal antibody targeting factor XI. We collected whole blood samples into two bags (each 240 ml, control group: enoxaparin 1.2 mg, treatment group: enoxaparin 1.2 mg plus abelacimab 5 mg) and circulated in a hemodialysis device for up to 3 h. We performed whole blood aggregation and thromboelastometry at several time points. Time to filter clotting was the primary endpoint. We included 10 volunteers. Each volunteer's blood was split into two bags (containing enoxaparin +/- abelacimab) and used simultaneously on two hemodialysis devices. The treatment group's time to filter clotting was significantly prolonged (treatment: 180 min, IQR 180-180 vs. control: 120 min, IQR 97-147, $p < 0.001$), and the transmembrane pressure was significantly lower at the end of the circuit flow (treatment: 13 mmHg vs. control: 65 mmHg, $p = 0.001$). Fibrinogen levels and median platelet counts were preserved. Platelet aggregation was better preserved in the treatment group for ristocetin ($p = 0.015$), thrombin receptor activating peptide ($p = 0.015$), and arachidonic acid ($p = 0.001$). Thromboelastometry showed prolonged clotting times in the treatment group at the end of the experiment (INTEM, $p < 0.001$; HEPTM, $p = 0.001$). Abelacimab prolonged the time to filter clotting in this ex vivo model of hemodialysis. This is an aggressive model due to the frequent re-circulation of blood and a lack of endothelial cells. These data provide support for testing abelacimab in patients on hemodialysis.

Pharmacokinetics and pharmacodynamics of Abelacimab (MAA868), a novel dual inhibitor of Factor XI and Factor XIa.

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Factor XI (FXI) inhibition offers the promise of hemostasis-sparing anticoagulation for the prevention and treatment of thromboembolic events. Abelacimab (MAA868) is a novel fully human monoclonal antibody that targets the catalytic domain and has dual activity against the inactive zymogen Factor XI and the activated FXI. To investigate the safety, pharmacokinetics (PK), and pharmacodynamics (PD) of single dose intravenous and multiple dose subcutaneous administration of abelacimab in healthy volunteers and patients with atrial fibrillation, respectively. In study ANT-003, healthy volunteers were administered single intravenous doses of abelacimab (30-150 mg) or placebo. The ANT-003 study also included a cohort of obese but otherwise healthy subjects. In study ANT-004, patients with atrial fibrillation were administered monthly subcutaneous doses of abelacimab (120 mg and 180 mg), or placebo, for 3 months. Key PK and PD parameters, including activated partial thromboplastin time (aPTT) and free FXI levels, as well as anti-drug antibodies (ADA) were assessed. Following intravenous administration of abelacimab, the terminal elimination half-life ranged from 25 to 30 days. One hour after the start of the intravenous infusion greater than 99% reductions in free FXI levels were observed. Following once monthly subcutaneous administration, marked reductions from baseline in free FXI levels were sustained. Parenteral administration of abelacimab demonstrated a favorable safety profile with no clinically relevant bleeding events. Intravenous and multiple subcutaneous dose administration of abelacimab were safe and well tolerated. The safety, PK, and PD data from these studies support the clinical development of abelacimab.

Abelacimab for Prevention of Venous Thromboembolism.

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The role of factor XI in the pathogenesis of postoperative venous thromboembolism is uncertain. Abelacimab is a monoclonal antibody that binds to factor XI and locks it in the zymogen (inactive precursor) conformation. In this open-label, parallel-group trial, we randomly assigned 412 patients who were undergoing total knee arthroplasty to receive one of three regimens of abelacimab (30 mg, 75 mg, or 150 mg) administered postoperatively in a single intravenous dose or to receive 40 mg of enoxaparin administered subcutaneously once daily. The primary efficacy outcome was venous thromboembolism, detected by mandatory venography of the leg involved in the operation or objective confirmation of symptomatic events. The principal safety outcome was a composite of major or clinically relevant nonmajor bleeding up to 30 days after surgery. Venous thromboembolism occurred in 13 of 102 patients (13%) in the 30-mg abelacimab group, 5 of 99 patients (5%) in the 75-mg abelacimab group, and 4 of 98 patients (4%) in the 150-mg abelacimab group, as compared with 22 of 101 patients (22%) in the enoxaparin group. The 30-mg abelacimab regimen was noninferior to enoxaparin, and the 75-mg and 150-mg abelacimab regimens were superior to enoxaparin ($P < 0.001$). Bleeding occurred in 2%, 2%, and none of the patients in the 30-mg, 75-mg, and 150-mg abelacimab groups, respectively, and in none of the patients in the enoxaparin group. This trial showed that factor XI is important for the development of postoperative venous thromboembolism. Factor XI inhibition with a single intravenous dose of abelacimab after total knee arthroplasty was effective for the prevention of venous thromboembolism and was associated with a low risk of bleeding. (Funded by Anthos Therapeutics; ANT-005 TKA EudraCT number, 2019-003756-37.).

News at XI: moving beyond factor Xa inhibitors.

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Oral anticoagulants are a mainstay for the prevention and treatment of arterial and venous thrombosis. Direct oral anticoagulants (DOACs) have replaced vitamin K antagonists for many indications. Currently available DOACs include dabigatran, which inhibits thrombin, and apixaban, edoxaban, and rivaroxaban, which inhibit factor (F) Xa. A new class of DOACs is under development. These new DOACs, which include asundexian and milvexian, inhibit FXIa, which is positioned in the intrinsic pathway of coagulation. Anticoagulants that target FXIa have the potential to be safer than the current DOACs because there is emerging evidence that FXI is essential for thrombosis but mostly dispensable for hemostasis. In addition to the oral inhibitors of FXIa, parenteral inhibitors are also under development. These include fesomersen, an antisense oligonucleotide that reduces the hepatic synthesis of FXI; abelacimab, an antibody that binds to FXI and blocks its activation; and osocimab, an FXIa inhibitory antibody. Focusing on these new agents, this article describes the unmet needs in oral anticoagulation therapy, explains why FXI is a promising target for new oral anticoagulants, reviews phase 2 clinical data on new agents, describes ongoing phase 3 trials, and provides a perspective on the opportunities and challenges for FXI inhibitors.

Factor XI, a potential target for anticoagulation therapy for venous thromboembolism.

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Venous thromboembolism (VTE) is a common cause of mortality and disability in hospitalized patients, and anticoagulation is an essential therapeutic option. Despite the increasing use of direct oral anticoagulants, complications and adverse drug reactions still occur in patients with VTE. Within 5 years, 20% of patients with VTE experience recurrence, and 50% of patients with deep vein thrombosis develop post-thrombotic syndrome. Furthermore, bleeding due to anticoagulants is a side effect that must be addressed. Therefore, safer and more effective anticoagulant strategies with higher patient compliance are urgently needed. Available epidemiological evidence and animal studies have shown that factor XI (FXI) inhibitors can reduce thrombus size and loosen the thrombus structure with a relatively low risk of bleeding, suggesting that FXI has an important role in thrombus stabilization and is a safer target for anticoagulation. Recent clinical trial data have also shown that FXI inhibitors are as effective as enoxaparin and apixaban in preventing VTE, but with a significantly lower incidence of bleeding. Furthermore, FXI inhibitors can be administered daily or monthly; therefore, the monitoring interval can be longer. Additionally, FXI inhibitors can prolong the activated partial thromboplastin time without affecting prothrombin time, which is an easy and common test used in clinical testing, providing a cost-effective monitoring routine for patients. Consequently, the inhibition of FXI may be an effective strategy for the prevention and treatment of VTE. Enormous progress has been made in the research strategies for FXI inhibitors, with abelacimab already in phase III clinical trials and most other inhibitors in phase I or II trials. In this review, we discuss the challenges of VTE therapy, briefly describe the structure and function of FXI, summarize the latest FXI/activated FXI (FXIa) inhibitor strategies, and summarize the latest developments in clinical trials of FXI/FXIa inhibitors.

Development of new anticoagulant in 2023: Prime time for anti-factor XI and XIa inhibitors.

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Thrombosis remains one of the leading causes of death in the world. The history of anticoagulation has evolved considerably from non-specific drugs (i.e., heparins and vitamin K antagonists, VKA) to agents that directly target specific coagulation factors (i.e., argatroban, fondaparinux and direct oral anticoagulants, DOAC). Since the last decade, DOAC are widely used in clinical practice because of their ease to use with favorable pharmacological profile and not requiring monitoring, particularly for venous thromboembolism treatment and prevention and stroke prevention in atrial fibrillation. However, despite having a better safety profile than VKA, their bleeding risk is not negligible. Therefore, research is underway to develop new anticoagulant therapies with a better safety profile. One of these new approaches to reduce the risk of bleeding is to target the coagulation in the intrinsic pathway, in particular the contact activation, with the ultimate goal of preventing thrombosis without impairing hemostasis. Based on epidemiological data with patients with inherited factor XI (FXI) deficiency and preclinical studies, FXI emerged as the most promising candidate target separating hemostasis from thrombosis. This review summarizes the role of FXI and FXIa in hemostasis, provides evidence of initial success with FXI pathway inhibitors in clinical trials (such as IONIS-FXIRx, fesomersen, osocimab, abelacimab, milvexian, asundexian or xisomab 3G3) and highlights the opportunities and challenges for this next generation of anticoagulants.